Reflect with us - Week 4

This assignment will not be graded and is only for practice.
Find the dimension of the vector space

$$V = \{(x, y, z) \mid x, y, z \in R, x + y + z = 0, z = 0\}$$
$$\text{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2); \quad (x_1, y_1, z_1), (x_2, y_2, z_2) \in V$$
$$\text{Scalar multiplication: } c(x, y, z) = (cx, cy, cz); \ (x, y, z) \in V, \ c \in R$$

V = {(x, y, z) ∣ x, y, z ∈ R, x + y + z = 0, z = 0}Addition: (x1, y1, z1) + (x2, y2, z2) = (x1 + x2, y1 + y2, z1 + z2); (x1, y1, z1), (x2, y2, z2) ∈ V Scalar multiplication: c(x, y, z) = (cx, cy, cz); (x, y, z) ∈ V , c ∈ R
Solution:Solution:

Observe that, in question, you are asked to find the dimension of the vector space . So first you have to find a basis of the vector space.
Note:Note: Consider the system of linear equations

$$\begin{aligned} x + y + z &= 0 \\ z &= 0 \end{aligned}$$
x + y + zz = 0 = 0

Find the solution of the system of linear equations.

Step 1:Step 1:

1 point
The system of linear equations has
- ◯ unique solution.
- ◯ no solution.
- ◯ infinitely many solution.

No, the answer is incorrect.
Score: 0
Feedback:
Use Gauss elimination method, and observe that $z = 0$z = 0 and $x = -y$.x = −y.
Accepted Answers:
infinitely many solution.

1 point
Is (1,-1,0) is a solution of the system of linear equations?
- ◯ Yes
- ◯ No

No, the answer is incorrect.
Score: 0
Feedback:
Substitute $x = 1, y = -1$x = 1, y = −1 and $z = 0$z = 0 in both the equations of system of linear equations and check.
Accepted Answers:
Yes

1 point
Is $(1, -1, 0) \in V$(1, −1, 0) ∈ V ?

○ Yes
○ No

No, the answer is incorrect.
Score: 0
Accepted Answers:
Yes

Claim: $\{(1,-1,0)\}$ spans V.

Step 2:

1 point

Let $(a,b,c) \in V$. Which of the following options is/are true?

☐ $a = -b$
☐ $a + b = 1$
☐ $c = 0$
☐ $c = 1$

No, the answer is incorrect.
Score: 0
Feedback:
As vector (a,b,c) is a vector of the vector space then it will satisfy the system of linear equations $a + b + c = 0, c = 0$

Accepted Answers:
$a = -b$
$c = 0$

Hence, any vector $(a,b,c) \in V$ can be written as $(a,-a,0)$.

1 point

Is $(a,-a,0)$ a scalar multiple of $(1,-1,0)$?

○ Yes
○ No

No, the answer is incorrect.
Score: 0
Accepted Answers:
Yes

1 point

If $(a,-a,0) = \alpha(1,-1,0)$, then what is the possible value for α?

○ a
○ $-a$
○ 0

No, the answer is incorrect.
Score: 0
Accepted Answers:
a

Hence, the set $\{(1, -1, 0)\}$$\{(1,-1,0)\}$ spans the vector space VV .

Step 3:Step 3: In this step we have to check whether the set $\{(1, -1, 0)\}$$\{(1,-1,0)\}$ is linearly independent or not.

1 point

Let SS be a set which contains only one non-zero vector. Then which of the following options is true?

○ SS is linearly independent set.

○ SS is linearly dependent set.

No, the answer is incorrect.
Score: 0
Accepted Answers:
SS is linearly independent set.

Recall:Recall: A basis of a vector space is a set of linearly independent vectors which spans the vector space.

From here you can conclude that $\{(1, -1, 0)\}$$\{(1,-1,0)\}$ is the basis of the vector space.

Step 4:Step 4: In this step we have to find the dimension the vector space VV .

Recall:Recall: The cardinality of the basis is the dimension of VV .
Find the dimension of the vector space VV .

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

Check Answers

Your score is: 0/8